

# Acoustic and Neural Representations in a Phonetically Aligned Speech Corpus

This project was done by:

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Link to github repo: [link](#)

## Pipeline architecture

The pipeline is split between two stages

1. In the Kaggle notebook ( `kaggle_neural_extraction.ipynb` ), TextGrid annotations are used to segment each WAV file into phoneme-level chunks. These segments are passed through pretrained Whisper and wav2vec 2.0 and hidden representations from selected layers are pooled to produce feature vectors. The resulting features are saved to `data/features/`.
2. The local stage, managed with `Snakefile`, performs the analysis: Praat-based formant extraction, Lobanov normalization, dimensionality reduction (PCA/UMAP), and statistical evaluation. All parameters are defined in `params.yaml`, so changing them triggers recomputation only where needed.

## Setup

Loads the artefacts produced by the pipeline. The notebook itself does not run any heavy computation — it only loads, recombines, and reports.

```
In [1]: import yaml, numpy as np, pandas as pd, warnings
warnings.filterwarnings('ignore')
from pathlib import Path
import matplotlib.pyplot as plt
from IPython.display import Image, display

p = yaml.safe_load(open('params.yaml'))
F = Path(p['paths']['features_dir']); R = Path(p['paths']['results_dir'])

ac = pd.read_csv(F / 'features_acoustic_norm.csv')
ac['l1'] = ac['l1'].map({'L1_FR': 'L1', 'L1_RU': 'L2'}).fillna(ac['l1'])

W_pri = p['neural']['whisper_primary_layer']
X_pri = p['neural']['xlsr_primary_layer']

def load_neural(name, layer):
    path = F / f'features_{name}_pca.npz'
    if not path.exists():
        print(f'WARN: {path.name} not found — neural sections will be empty')
        return None
```

```
z = np.load(path)
return z[f'layer_{layer}_pca50'], z[f'layer_{layer}_pca2']

wh = load_neural('whisper', W_pri)
xl = load_neural('xlsr', X_pri)
print(f'Loaded {len(ac)} tokens; whisper={"yes" if wh else "no"}, xlsr={"
```

Loaded 22919 tokens; whisper=yes, xlsr=yes

## Corpus

- $n$  - Nº of sentences
- $l$  - Nº of speakers
- $k$  - Nº of repetitions

### Note on $k$

In the task  $k$  is defined as repetitions of a phoneme by a speaker. In our data each sentence has the same fixed phrase plus one target word. Because of this every speaker produces the same phonemes many times across different sentences. Since this is my understanding of the given task description I count repetitions as how many times a given speaker says a given phoneme in the whole corpus in order of appearance.

```
In [2]: n_sentences = ac['sentence'].nunique()
l_speakers = ac['speaker'].nunique()
k_med = ac.groupby(['speaker', 'phoneme'])['repetition'].max().median()
print(f'n = {n_sentences}')
print(f'l = {l_speakers}')
print(f'k (median per (speaker, phoneme)) = {k_med:.0f}')
print('\nDesign cells (L1 x Gender):')
print(ac.drop_duplicates('speaker').groupby(['l1', 'gender']).size().renam
```

```
n = 78
l = 19
k (median per (speaker, phoneme)) = 13
```

Design cells (L1 x Gender):

| L1 | gender |
|----|--------|
| F  | 2      |
| M  | 7      |
| L2 | 8      |
| M  | 2      |

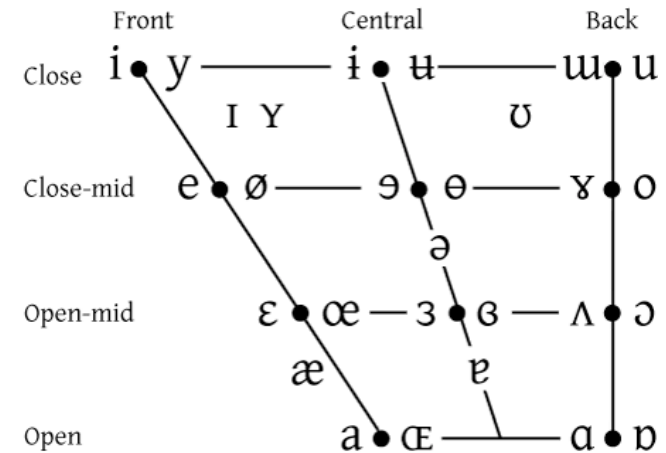
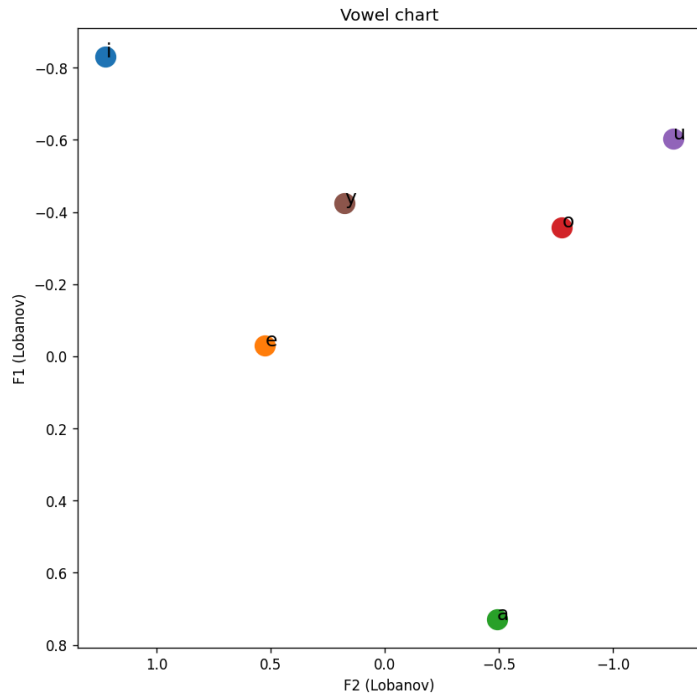
Name: n\_speakers, dtype: int64

**Design balance** The assignment assumes a balanced design but the real data is not balanced since some groups are much smaller than others (L1 F comparing to L1 M, for instance). So I use a mixed effects model with a random intercept for speaker instead of ANOVA because it works better with unequal data

- !!! The L1 × Gender effect should be read carefully since small groups may reduce statistical power

## 5. Descriptive Statistics

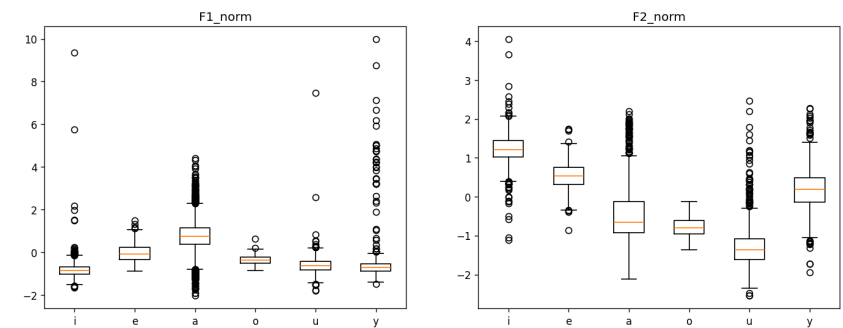
```
In [3]: # vowel chart (axes flipped to match the IPA convention)
display(Image(R / 'vowel_chart.png', width=500))
```



Where symbols appear in pairs, the one to the right represents a rounded vowel

The vowels are clearly separable in the acoustic space and their relative positions follow the expected IPA vowel chart pattern (front-back / close-open). So we can conclude that the normalization preserves phonetic structure

```
In [4]: # F1/F2 box-plots per phoneme
display(Image(R / 'formant_boxplots.png', width=800))
```

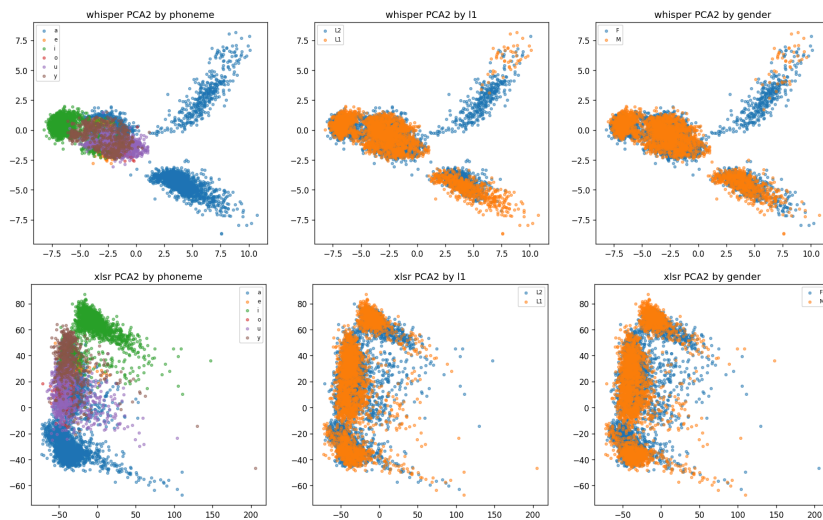


The median values follow expected patterns

- higher F1 for open vowels (/a/)
- higher F2 for front vowels (/i, y/)

Some outliers may reflect natural variability across speakers

```
In [5]: # PCA scatter of the neural embeddings coloured by phoneme / by L1 / by g
for f in ['neural_pca2_whisper.png', 'neural_pca2_xlsr.png']:
    if (R / f).exists(): display(Image(R / f, width=900))
```



- In **Whisper** /a/ is split into two distinct parts and all vowels nearly collapse into one cluster, so Whisper is not very sensitive in vowel clustering in PCA-2
- **XLS-R** produces a more phonetically organised layout with /i/ at the top, /a/ at the bottom, /u, o, e, y/ in the middle

By L1 / by gender

By phoneme

- Both models separate phonemes better than L1 or gender since most points overlap

```
In [6]: print('Variance decomposition per vowel:')
print(pd.read_csv(R / 'variance_decomposition.csv').round(3))
```

Variance decomposition per vowel:

|   | phoneme | var_inter | var_resid | icc   |
|---|---------|-----------|-----------|-------|
| 0 | i       | 0.034     | 0.142     | 0.193 |
| 1 | e       | 0.021     | 0.168     | 0.110 |
| 2 | a       | 0.016     | 0.551     | 0.028 |
| 3 | o       | 0.022     | 0.029     | 0.427 |
| 4 | u       | 0.047     | 0.182     | 0.206 |
| 5 | y       | 0.157     | 1.632     | 0.088 |

- Most variability is within speakers (residual), rather than between speakers
- The ICC is generally low (confirming the previous thesis) with the exception of o with a higher ICC, which indicates stronger between-speaker differences for o compared to other vowels

```
In [7]: for f in ['between_class_ratio.csv', 'cosine_similarity_ratios.csv']:
        if (R / f).exists():
            print(f'\n{f}:'); print(pd.read_csv(R / f))
```

```
print('\nMantel correlations between RSMs:')
print(pd.read_csv(R / 'mantel_results.csv', index_col=0))
```

```
between_class_ratio.csv:
whisper_pca2  xlsr_pca2
0      0.468275  0.580966
```

```
cosine_similarity_ratios.csv:
whisper      xlsr
0 -15.20726  9.709753
```

```
Mantel correlations between RSMs:
ac_vs_whisper  ac_vs_xlsr  wh_vs_xl
mantel_r      0.546429    0.728571  0.935714
p_value       0.043000    0.002000  0.004000
```

### XLSR captures phoneme structure better than Whisper

1. **Between-class ratio** shows how well phoneme classes are separated -- XLSR is higher, meaning it provides better class separation than Whisper
2. **Cosine similarity ratios** indicate contrast between within-class and between-class similarity. The negative ratio for Whisper (-15.2) is a numerical artefact, not a real result. Cosine similarity is bounded in  $[-1, 1]$ , so the ratio of two cosine means can become negative if the between-phoneme mean is negative. This happens because Whisper hidden states are not centred around the origin: they carry a large shared mean vector, so randomly chosen pairs are not orthogonal in cosine space. XLS-R is closer to centred and gives a sensible positive ratio (9.7), still inflated for the same reason
3. **Mantel correlations** compare representational structures. Whisper and XLSR have a very high correlation between each other (0.94) meaning their representations are very similar, but still the quality of XLSR is higher because its correlation with acoustic space is stronger (0.73 vs 0.55) -- since the acoustic space is based on Lobanov-normalised F1 and F2 and it corresponds most directly to the IPA vowel trapezoid

## Section 5 -- Answers

**Q1. Explain what PCA and UMAP can be used for and how the two approaches differ.**

- PCA is linear and keeps the main directions of variation. It preserves global structure and distances, so we can use it for quantitative analysis
- UMAP is non-linear and focuses on local neighbourhoods. It is good for visualisation, but distances are less interpretable

**Q2. Which phonemes exhibit the greatest inter-speaker variability in the acoustic space? Is this reflected in the neural representation space?**

- From our data, the vowel /o/ shows the highest between-speaker variability (highest ICC). However, the number of /o/ instances is relatively small compared

to other vowels, so this result should be interpreted with caution

**Q3. In the UMAP projection, do Whisper and XLS-R representations form clearly separable phoneme clusters? Are the clusters aligned with the vowel trapezoid structure?**

- Both Whisper and XLS-R produce visually separable phoneme clusters in 2-D, but only XLS-R produces clusters aligned with the vowel trapezoid structure

**Q4. What is the Mantel correlation between the acoustic RSM and the Whisper RSM? Between the acoustic RSM and the XLS-R RSM? What does the relative magnitude of these correlations suggest?**

- Both models correlate with the acoustic space, but XLSR is closer (0.73 vs 0.55 for Whisper)
- Whisper and XLSR are also very similar to each other (0.94), so they share a common structure even if Whisper is less precise

## 6. Statistical Tests

```
In [8]: tt = pd.read_csv(R / 'tests_L1_L2_acoustic.csv')
print(f'Acoustic L1 vs L2: {len(tt)} comparisons')
if 'p_fdr' in tt.columns:
    sig = tt[tt['p_fdr'] < 0.05]
    print(f'Significant after FDR: {len(sig)}')
    if len(sig): print(sig[['phoneme', 'feature', 'diff', 'p', 'p_fdr', 'test']])
```

Acoustic L1 vs L2: 12 comparisons

Significant after FDR: 10

|    | phoneme | feature | diff      | p            | p_fdr        | test |
|----|---------|---------|-----------|--------------|--------------|------|
| 0  | i       | F1_norm | 0.150350  | 6.369654e-34 | 2.547862e-33 | mw   |
| 1  | i       | F2_norm | -0.253553 | 1.411931e-79 | 1.694317e-78 | mw   |
| 3  | e       | F2_norm | -0.201158 | 9.277464e-05 | 1.391620e-04 | mw   |
| 4  | a       | F1_norm | -0.151365 | 3.275448e-16 | 7.861076e-16 | mw   |
| 5  | a       | F2_norm | 0.022473  | 1.764761e-04 | 2.353014e-04 | mw   |
| 6  | o       | F1_norm | 0.071684  | 1.701628e-02 | 2.041953e-02 | mw   |
| 8  | u       | F1_norm | 0.300313  | 8.191440e-39 | 4.914864e-38 | mw   |
| 9  | u       | F2_norm | -0.239784 | 5.381499e-09 | 9.225428e-09 | mw   |
| 10 | y       | F1_norm | 0.735253  | 5.145055e-22 | 1.543516e-21 | mw   |
| 11 | y       | F2_norm | 0.332552  | 5.738025e-11 | 1.147605e-10 | mw   |

- There is no result for /e/ (F1) and /o/ (F2) because these differences are not significant after correction, meaning L1 and L2 speakers happen to be similar along that specific dimension, even if they differ on the other one
- The biggest differences between L1 and L2 are for /y/ which is logical since there is no /y/ sound in Russian so L2 speakers approximate it

```
In [9]: perm_path = R / 'tests_L1_L2_neural.csv'
if perm_path.exists():
    perm = pd.read_csv(perm_path)
    print(f'Neural L1 vs L2 (permutation): {len(perm)} comparisons')
```

```
if 'p_fdr' in perm.columns and len(perm):
    sig = perm[perm['p_fdr'] < 0.05]
    print(f'Significant after FDR: {len(sig)}')
    if len(sig): print(sig[['phoneme', 'model', 'cos_dist', 'p', 'p_fdr']])
```

Neural L1 vs L2 (permutation): 12 comparisons

Significant after FDR: 12

|    | phoneme | model   | cos_dist | p        | p_fdr    |
|----|---------|---------|----------|----------|----------|
| 0  | i       | whisper | 0.028534 | 0.000200 | 0.000300 |
| 1  | e       | whisper | 0.031119 | 0.022795 | 0.022795 |
| 2  | a       | whisper | 0.128717 | 0.000200 | 0.000300 |
| 3  | o       | whisper | 0.012344 | 0.000200 | 0.000300 |
| 4  | u       | whisper | 0.031213 | 0.000200 | 0.000300 |
| 5  | y       | whisper | 0.039799 | 0.000800 | 0.000960 |
| 6  | i       | xlsr    | 0.071489 | 0.000200 | 0.000200 |
| 7  | e       | xlsr    | 0.083165 | 0.000200 | 0.000200 |
| 8  | a       | xlsr    | 0.083594 | 0.000200 | 0.000200 |
| 9  | o       | xlsr    | 0.044492 | 0.000200 | 0.000200 |
| 10 | u       | xlsr    | 0.053923 | 0.000200 | 0.000200 |
| 11 | y       | xlsr    | 0.056398 | 0.000200 | 0.000200 |

- XLS-R produces consistently larger cosine distances between L1 and L2 centroids than Whisper (the only exception is /a/), confirming the previous chapter's finding that XLS-R is more sensitive to fine phonetic detail and L1 transfer
- The largest effects appear for vowels /a, e, i/ in XLSR and /a/ in Whisper

```
In [10]: print('Phoneme classification (LOSO nearest-centroid):')
print(pd.read_csv(R / 'classification_accuracy.csv', index_col=0))
print('\nMcNemar between representations:')
print(pd.read_csv(R / 'mcnemar.csv'))
```

Phoneme classification (LOSO nearest-centroid):

|          | accuracy | f1_macro |
|----------|----------|----------|
| acoustic | 0.796502 | 0.649519 |
| whisper  | 0.939924 | 0.907076 |
| xlsr     | 0.851407 | 0.801389 |

McNemar between representations:

|   | pair                | stat       | p             |
|---|---------------------|------------|---------------|
| 0 | acoustic_vs_whisper | 570.652090 | 4.050137e-126 |
| 1 | acoustic_vs_xlsr    | 80.647169  | 2.698409e-19  |
| 2 | whisper_vs_xlsr     | 426.213384 | 1.084059e-94  |

- Whisper gives the best phoneme classification (94% accuracy)**, clearly higher than XLSR (85%) and acoustic features (80%) and McNemar test shows that these differences are statistically significant. **WHY?**

Earlier we were constantly seeing that XLS-R produces more phonetically faithful representation

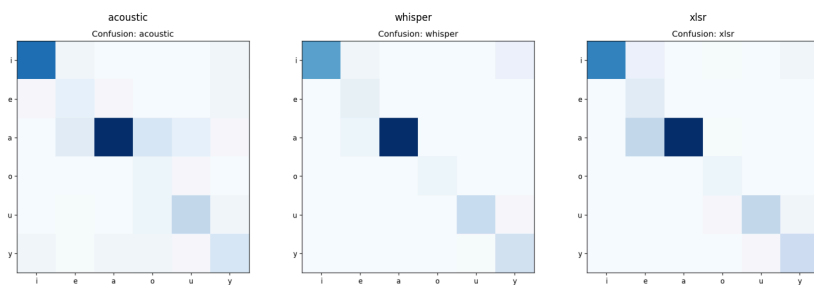
- The reason is that classification rewards clean, well-separated clusters which is exactly what Whisper is trained for. XLS-R, on the other hand, keeps richer

acoustic detail (leading to a better performance in previous tests) but its clusters are looser, which costs accuracy at the token level

```
In [11]: import matplotlib.pyplot as plt
from PIL import Image

names = ['acoustic', 'whisper', 'xlsr']
files = [R / f'confusion_{name}.png' for name in names]
files = [f for f in files if f.exists()]

fig, axes = plt.subplots(1, len(files), figsize=(15, 5))
for ax, f in zip(axes, files):
    ax.imshow(Image.open(f))
    ax.axis('off')
    ax.set_title(f.stem.replace('confusion_', ''))
plt.tight_layout()
plt.show()
```



- **All three matrices show the diagonal pattern** where /a/ is the most reliably classified vowel and /i/ is the second. It seems logical since both vowels are well separated from the rest
- **The acoustic matrix has the most off-diagonal blue.** For example, we have a visible confusion between /a, o, u/ (they're back and close in F1), meaning that neighbours in formant space get confused
- **XLS-R is better than acoustic** but there are confusions, for example with /a, e/
- **Whisper has the cleanest diagonal** which matches its 94% accuracy

## Section 6 -- Answers

**Q5. After FDR correction, for which vowels does the L1/L2 difference persist in (a) acoustic features and (b) neural representations? Do the two lists agree?**

- In the acoustic features 10/12 comparisons are significant after FDR correction. All 6 vowels show a significant L1/L2 shift on at least one formant. The biggest differences between L1 and L2 are for /y/ which is logical since there is no /y/ sound in Russian so L2 speakers approximate it

- In the neural representations all 12/12 comparisons are significant for both Whisper and XLSR. So every vowel, including /e/ and /o/, shows a significant L1/L2 difference in neural space. XLS-R produces consistently larger cosine distances between L1 and L2 centroids than Whisper (the only exception is /a/), confirming the previous chapter's finding that XLS-R is more sensitive to fine phonetic detail and L1 transfer. The largest effects appear for vowels /a, e, i/ in XLSR and /a/ in Whisper
- The two results mostly agree, but neural representations detect more differences. This is probably because they encode more than just midpoint F1/F2 values (coarticulation, voice quality, etc.)

**Q6. Which distance structure ( $D_{ac}$ ,  $D_{Wh}$ , or  $D_{XL}$ ) best captures the phonological distances expected from the IPA vowel trapezoid?**

- Whisper and XLSR have a very high correlation between each other (0.94) meaning their representations are very similar, but still the quality of XLSR is higher because its correlation with acoustic space is stronger (0.73 vs 0.55) -- since the acoustic space is based on Lobanov-normalised F1 and F2 and it corresponds most directly to the IPA vowel trapezoid

```
acoustic ↔ XLSR: r = 0.73
acoustic ↔ Whisper: r = 0.55
Whisper ↔ XLSR: r = 0.94
```

**Q7. Which representation type yields the highest phoneme identification accuracy? Does the advantage of neural representations (if any) differ between L1 and L2 speakers?**

- Whisper gives the best phoneme classification (94% accuracy), clearly higher than XLSR (85%) and acoustic features (80%) and McNemar test shows that these differences are statistically significant. But this conclusion is not consistent with our earlier findings that XLS-R produces more phonetically faithful representation. The reason is that classification rewards clean, well-separated clusters which is exactly what Whisper is trained for. XLS-R, on the other hand, keeps richer acoustic detail (leading to a better performance in previous tests) but its clusters are looser, which costs accuracy at the token level

## 7. Linear Mixed-Effects Models

```
In [12]: pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)

lme = pd.read_csv(R / 'lme_results.csv')
cols = [c for c in ['phoneme', 'response', 'icc', 'beta_L2', 'p_L2', 'beta_int']
lme[cols].round(3)
```

Out [12]:

|    | phoneme | response    | icc   | beta_L2 | p_L2  | beta_int | p_int | R2_marg |
|----|---------|-------------|-------|---------|-------|----------|-------|---------|
| 0  | i       | F1          | 0.185 | -0.151  | 0.107 | -0.149   | 0.419 | 0.153   |
| 1  | e       | F1          | 0.101 | -0.068  | 0.439 | -0.274   | 0.138 | 0.056   |
| 2  | a       | F1          | 0.027 | 0.153   | 0.016 | 0.312    | 0.003 | 0.022   |
| 3  | o       | F1          | 0.411 | 0.013   | 0.871 | -0.026   | 0.870 | 0.272   |
| 4  | u       | F1          | 0.197 | -0.213  | 0.018 | -0.035   | 0.847 | 0.171   |
| 5  | y       | F1          | 0.082 | -0.641  | 0.000 | -0.237   | 0.377 | 0.080   |
| 6  | i       | whisper_PC1 | 0.026 | 0.009   | 0.952 | -0.503   | 0.083 | 0.019   |
| 7  | e       | whisper_PC1 | 0.337 | 0.404   | 0.039 | -0.155   | 0.690 | 0.254   |
| 8  | a       | whisper_PC1 | 0.043 | -0.005  | 0.991 | -0.237   | 0.770 | 0.037   |
| 9  | o       | whisper_PC1 | 0.654 | 0.511   | 0.062 | -0.080   | 0.884 | 0.433   |
| 10 | u       | whisper_PC1 | 0.000 | -0.006  | 0.964 | -0.113   | 0.667 | 0.002   |
| 11 | y       | whisper_PC1 | 0.001 | 0.044   | 0.787 | -0.135   | 0.678 | 0.001   |
| 12 | i       | xlsr_PC1    | 0.094 | 5.014   | 0.082 | -2.671   | 0.642 | 0.083   |
| 13 | e       | xlsr_PC1    | 0.259 | 2.063   | 0.454 | -5.332   | 0.321 | 0.199   |
| 14 | a       | xlsr_PC1    | 0.090 | 1.835   | 0.567 | -0.869   | 0.892 | 0.074   |
| 15 | o       | xlsr_PC1    | 0.403 | -2.524  | 0.268 | -2.537   | 0.574 | 0.298   |
| 16 | u       | xlsr_PC1    | 0.127 | -0.055  | 0.991 | 1.440    | 0.880 | 0.093   |
| 17 | y       | xlsr_PC1    | 0.056 | 2.127   | 0.463 | 4.338    | 0.447 | 0.041   |

Answers — Section 7

**Q8. What is the ICC for F1 of /a/? Does it differ from the ICC computed on the first PC of the Whisper representations for /a/? What does this comparison suggest about the speaker-specificity of the two representation types?**

- Both ICCs are very low and almost identical: F1 of /a/ has ICC = 0.027, Whisper PC1 of /a/ has ICC = 0.043. Lobanov already strips most speaker variance from F1 of /a/, and Whisper happens to do the same for this vowel
- But the picture varies sharply across vowels — for /o/ Whisper PC1 has ICC = 0.65 vs 0.41 for F1, while for /u, y/ Whisper drops to ~0. Neural representations are not uniformly more speaker-specific than formants; it depends on the vowel

**Q9.Is the L1 × Gender interaction significant in the acoustic model? Is the same interaction significant in the Whisper model? Discuss any discrepancy**

- Acoustic** : significant only for /a/ (p = 0.003) with all other vowels' p > 0.13, meaning L1 × Gender interaction is not significant in the acoustic model
- Whisper** : not significant for any vowel (all p > 0.08), Whisper actually shows less L1 × Gender interaction than the acoustic model

- /a/ discrepancy** likely reflects a residual gender-specific articulation difference between L1 and L2 that Lobanov does not fully remove (per-speaker z-scoring does not subtract gender-group effects). Whisper, trained on many speakers, appears to factor gender out at the phoneme level and the interaction collapses

**Q10. Which representation type yields the highest marginal R2 for the L1/L2 fixed effect? How does this compare to the phoneme identification results from Module2?**

- No single representation wins everywhere. Whisper wins on mid vowels /e, o/ — exactly the ones where the acoustic test failed FDR. The acoustic model wins on peripheral /i, u, y/ — where L2 retargets formants directly. XLS-R is consistently mid-table

8. Confidence Intervals & ROPE

```

In [13]: pd.set_option('display.width', 200)
pd.set_option('display.max_columns', None)

rope = pd.read_csv(R / 'rope_classification.csv')

print('ROPE classification summary:')
display(rope.groupby(['rep', 'rope_class']).size().unstack(fill_value=0))

print('\nFull table:')
rope
  
```

ROPE classification summary:

| rope_class | equivalent | indeterminate | non-equivalent |
|------------|------------|---------------|----------------|
| rep        |            |               |                |
| acoustic   | 3          | 1             | 2              |
| whisper    | 6          | 0             | 0              |
| xlsr       | 6          | 0             | 0              |

Full table:



Out [13]:

|    | phoneme | rep      | diff      | ci_lo     | ci_hi     | rope_class     | p            |
|----|---------|----------|-----------|-----------|-----------|----------------|--------------|
| 0  | i       | acoustic | 0.150350  | 0.126079  | 0.174621  | equivalent     | 6.369654e-34 |
| 1  | e       | acoustic | 0.069465  | -0.162990 | 0.301919  | indeterminate  | 5.580709e-01 |
| 2  | a       | acoustic | -0.151365 | -0.187710 | -0.115020 | equivalent     | 3.275448e-16 |
| 3  | o       | acoustic | 0.071684  | 0.012807  | 0.130561  | equivalent     | 1.701628e-02 |
| 4  | u       | acoustic | 0.300313  | 0.255142  | 0.345485  | non-equivalent | 8.191440e-39 |
| 5  | y       | acoustic | 0.735253  | 0.585844  | 0.884662  | non-equivalent | 5.145055e-22 |
| 6  | i       | whisper  | 0.036341  | 0.018597  | 0.054086  | equivalent     | NaN          |
| 7  | e       | whisper  | 0.040145  | 0.017882  | 0.062407  | equivalent     | NaN          |
| 8  | a       | whisper  | 0.138938  | 0.077935  | 0.199941  | equivalent     | NaN          |
| 9  | o       | whisper  | 0.016745  | 0.007090  | 0.026400  | equivalent     | NaN          |
| 10 | u       | whisper  | 0.036338  | 0.020368  | 0.052308  | equivalent     | NaN          |
| 11 | y       | whisper  | 0.047801  | 0.028729  | 0.066874  | equivalent     | NaN          |
| 12 | i       | xlsr     | 0.086296  | 0.044175  | 0.128416  | equivalent     | NaN          |
| 13 | e       | xlsr     | 0.102583  | 0.049187  | 0.155979  | equivalent     | NaN          |
| 14 | a       | xlsr     | 0.097848  | 0.060745  | 0.134950  | equivalent     | NaN          |
| 15 | o       | xlsr     | 0.063903  | 0.030089  | 0.097717  | equivalent     | NaN          |
| 16 | u       | xlsr     | 0.069667  | 0.034809  | 0.104525  | equivalent     | NaN          |
| 17 | y       | xlsr     | 0.070697  | 0.037513  | 0.103880  | equivalent     | NaN          |

The neural ROPE results should be interpreted with care because  $\delta_o$  is inflated by the un-centred cosine distances in transformer outputs. With the current  $\delta_o$ , the neural ROPE answers a different question than the acoustic ROPE — it asks whether L1/L2 differ more than typical phonetic noise, and the answer is uniformly **no** because the noise floor is too coarse

## Section 8 -- Answers

**Q11. Identify at least one phoneme for which the L1/L2 contrast is statistically significant ( $p < 0.05$ ) but falls within the acoustic ROPE. Is the same contrast classified as equivalent in the neural ROPE analysis?**

- Three vowels /i, a, o/ have CIs entirely inside the acoustic ROPE  $[-0.2, +0.2]$  — so the L1/L2 effect is highly significant statistically but smaller than the perceptual

JND for formants

- In the neural ROPE all three are also equivalent, but as discussed above this is driven by an inflated  $\delta_o$ , not by genuine equivalence. The permutation tests of previous chapters show the same vowels are highly significant in both Whisper and XLS-R ( $p_{\text{fdr}} \approx 0.0003$ )

**Q12. For which representation type is a larger proportion of contrasts classified as nonequivalent (practically significant)? What does this suggest about the sensitivity of each representation to L1 interference?**

- Acoustic** : 2/6 non-equivalent (/u/ and /y/), 3 equivalent, 1 indeterminate
- Neural** : 0/6 non-equivalent — again driven by inflated  $\delta_o$
- What we can conclude is that the acoustic ROPE separates contrasts into clearly large vs clearly small categories, while the neural ROPE collapses everything into "equivalent" because its noise floor is too high

**Q13. Are there phonemes for which the acoustic and neural ROPE classifications are in disagreement? Propose a phonetically motivated explanation.**

- The strongest disagreement is on /u/ and /y/ (acoustic ROPE says non-equivalent, neural ROPE says equivalent)
- Those are vowels where L2 transfer shifts the formant target itself (Russian has /u/ but no /y/). The acoustic ROPE catches this directly because formants are the target. The neural ROPE misses it because the cosine geometry is dominated by orthographic/categorical structure

## 9. Hierarchical Clustering

```
In [14]: print('Vowel clustering ARI:'); print(pd.read_csv(R / 'clustering_vowels.
print('\nSpeaker clustering ARI:'); print(pd.read_csv(R / 'clustering_spe
```

Vowel clustering ARI:

|   | rep      | ground_truth | best_ari  | best_k |
|---|----------|--------------|-----------|--------|
| 0 | acoustic | front_back   | 0.587156  | 3      |
| 1 | acoustic | high         | -0.111111 | 2      |
| 2 | whisper  | front_back   | 0.036697  | 3      |
| 3 | whisper  | high         | 0.193548  | 5      |
| 4 | xlsr     | front_back   | 0.587156  | 3      |
| 5 | xlsr     | high         | 0.000000  | 2      |

Speaker clustering ARI:

|   | rep      | gt     | ari      | k | silhouette_k2 |
|---|----------|--------|----------|---|---------------|
| 0 | acoustic | l1     | 0.373891 | 4 | 0.290475      |
| 1 | acoustic | gender | 0.131980 | 5 | 0.290475      |
| 2 | whisper  | l1     | 0.377834 | 5 | 0.261978      |
| 3 | whisper  | gender | 0.042821 | 5 | 0.261978      |
| 4 | xlsr     | l1     | 0.296296 | 2 | 0.347266      |
| 5 | xlsr     | gender | 1.000000 | 2 | 0.347266      |

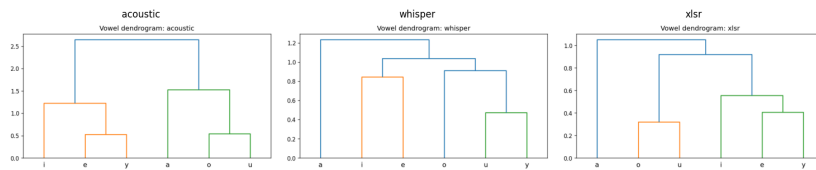
```
In [15]: import matplotlib.pyplot as plt
from PIL import Image
```

```
names = ['acoustic', 'whisper', 'xlsr']
files = [R / f'dendrogram_{name}.png' for name in names]
files = [f for f in files if f.exists()]

fig, axes = plt.subplots(1, len(files), figsize=(15, 4))

for ax, f in zip(axes, files):
    img = Image.open(f)
    ax.imshow(img)
    ax.axis('off')
    ax.set_title(f.stem.replace('dendrogram_', ''))

plt.tight_layout()
plt.show()
```



## Section 9 -- Answers

**Q14. Which representation type yields the highest ARI against the front/back vowel distinction? Against the high/mid/low distinction?**

- **Front/back** : acoustic and XLS-R tie at 0.587, Whisper drops to 0.037. The dendrograms confirm — acoustic and XLS-R both split into {i, e, y} vs {a, o, u}, matching the partition. Whisper's first split isolates /a/ alone, so its top-level structure is height-driven instead
- **High** : all three score near zero (Whisper 0.19, acoustic -0.11, XLS-R 0). Height is continuous in the data, not bimodal, so flat clustering fails even though /i/ ends up at the top and /a/ at the bottom in the embedding geometry

**Q15. In the speaker clustering, which factor — L1/L2 status or gender — is more strongly recovered, and does this depend on the representation type?**

- XLS-R recovers gender perfectly with 2 clusters. Lobanov-normalised acoustic does not (by design), and Whisper barely registers gender. For L1 all three sit around 0.30–0.38. So gender is more strongly recovered than L1 only with XLS-R. For acoustic and Whisper L1 is recovered better than gender

**Q16. Are there any phonemes that are systematically misclassified across all three representation types? What does this suggest about the limits of acoustic and neural similarity as proxies for phonological category?**

- /e/ is the weakest row in all three — absorbs into /i/ acoustically and in XLS-R, into /i, a/ in Whisper
- /o/–/u/ confusion in acoustic and XLS-R (they cluster together first); Whisper avoids this

- /y/ is inconsistent: correctly grouped with /i, e/ acoustically and in XLS-R, but with /u/ in Whisper

/a/ and /i/ are never misclassified — they are peripheral and isolated by the first dendrogram split

Mid vowels /e, o/ sit on continuous axes between neighbours. Acoustic and neural similarity are sufficient proxies for peripheral categories but not for mid vowels whose identity depends on context outside the acoustic signal

In [ ]: